

The Metabolic Interpretation of FSCTF: Eating as the Creation Cycle

Paper V in the FSCTF series

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Code: `genesis-simulation/dynamics/metabolic.js`,
`genesis-simulation/tests/test-metabolic-theory.mjs` (76 tests, zero failures).

Abstract

The FSCTF creation cycle $R = G(W(L(\Omega)))$ admits a precise metabolic reading: every digestion is the equation running on external matter with the witness as invariant. This paper makes the reading mathematical. We show that the four phases of metabolism (ingestion, digestion, absorption, excretion) correspond bijectively to four algebraic operations in the Clifford algebra $\text{Cl}(3,1)$, that the digestive seam is the Peirce cross-sector of a primitive idempotent split, and that the difference between devouring and communion is parameterized by a single integer (the number of Grace contraction steps). Eleven theorems, each independently verifiable by short numerical script, establish the framework rigorously: seven core theorems on the metabolic cycle plus four pathology theorems (autoimmunity, immunodeficiency, microbiome, constipation) that originated as biological predictions and are here promoted to pure algebra. The paper makes no claims to novelty in the underlying algebra: everything reduces to operations already present in the substrate of Paper I [1]. The contribution is the bijection, the identification of two distinct witness invariants, the closed-form decay law $c_k(t) = c_k(0) \exp(-t(1 - \varphi^{-k}))$ for grade- k Grace dynamics, and four algebraic pathology theorems with explicit biological readings.

Three load-bearing results.

(M1) The metabolic cycle is the creation cycle. (*theorems 3.2 and 3.3.*) The map

$$(\text{ingest, digest, absorb+metabolize, excrete}) \longleftrightarrow (M + \lambda I, \text{decompose}, G^n, \text{mixed-reconstruct})$$

is a bijection of operational primitives in $\text{Cl}(3,1)$. The composition satisfies energy conservation absorbed + residue = mixed exactly.

(M2) Grace decay is closed-form grade-diagonal. (*theorem 3.4.*) Under the Grace contraction $G : \mathbb{R}^{16} \rightarrow \mathbb{R}^{16}$ given by $G(c)_i = \varphi^{-\text{grade}(e_i)} c_i$ in any orthonormal grade basis,

$$c_k(t) = c_k(0) \cdot \exp(-t(1 - \varphi^{-k})), \quad k \in \{0, 1, 2, 3, 4\}.$$

The grade-0 (scalar witness) coefficient is the unique fixed point; the pseudoscalar (grade 4) decays fastest. Verified to 10^{-10} .

(M3) Two distinct witness invariants. (*theorem 4.1.*) The scalar coefficient c_0 is the eigenmode-1 of `graceContract` in coefficient space, while the matrix row $M[0, :]$ is the eigenmode-1 of `masterStep` in matrix space. *Neither preserves the other's invariant.* This distinction was implicit in the existing simulation code and is made explicit here. It resolves an ambiguity in `EverythingFromWitness` §4.

Scope. All proofs reduce to closed-form algebra in $\text{Cl}(3, 1) \cong M_4(\mathbb{R})$. The framework is a re-reading, not a new physics; its empirical content lies in specific biological predictions stated in §9.

What this paper does.

- Reads the FSCTF creation cycle $R = G(W(L(\Omega)))$ as a *metabolic* interpretation of the substrate: opening Grace, decomposing food (Sacrifice), retaining the boundary-preserving residue.
- Encodes healthy and pathological eating as accepted vs. rejected sequences on a 64-state braid-grammar finite-state transducer over the alphabet $\{W, G, S, C, L, P, X\}$; proves the accepting depths are exactly $\{2, 3, 4, 7\}$.
- Derives four *pathology theorems* – autoimmunity, immunodeficiency, microbiome transport, constipation – as failure modes of the same boundary algebra that underwrites the substrate.

Place in the series. This is the *operational/biological* side paper for the main trilogy ($\text{I} \rightarrow \text{II} \rightarrow \text{III}$). It depends on Paper I [1] (Clifford algebra, Peirce idempotents) and Paper III [3] (monad/braid grammar) but is independent of Papers II, VI, VII, and VIII.

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1 Introduction

This paper takes a sentence from natural language — “eating is asymmetric coherence transfer across a boundary” — and shows that it has a unique algebraic realization in the substrate of Paper I [1]. The realization makes seven testable numerical claims, all of which we verify, and four biological predictions, which we state but do not verify.

The question

What is eating?

The biological answer (a system imports matter, decomposes it, absorbs compatible structure, rejects incompatible residue) is operational but underdetermines the formalism. We ask: under which choice of algebraic primitives does the operational answer become unique?

The answer, given a Clifford algebra $\text{Cl}(3, 1)$ over $\mathbb{R}$ with primitive idempotents and a contraction operator G satisfying the Lyapunov property of Paper I [1], is exactly one. The bijection is forced by:

- (1) Energy conservation across the cycle (forces excretion as exact residue, not as decay).
- (2) Witness invariance through ingestion (forces absorption to be a Grace-stable projection).
- (3) Boundary-status (not geometric) interior/exterior (forces the digestive surface to be the Peirce cross-sector of an idempotent split, not a geometric boundary).
- (4) Single-parameter dial between devouring and communion (forces the parameter to be the integer Grace step count n).

The bijection

Metabolic phase	Algebraic operation
ingestion	$\text{mixed} = \text{self} + \lambda \cdot \text{input}$
digestion	$c = \text{decompose}(\text{mixed})$
absorption + metabolism	iterated Grace: $c \mapsto G^n(c)$
excretion	$\text{residue} = \text{mixed} - \text{reconstruct}(G^n(c))$

Relation to Papers I–IV

This paper depends on Paper I [1] for the algebraic substrate ($\text{Cl}(3, 1)$, the Grace contraction, the idempotent split). It does not depend on Papers II–IV [2, 3, 4]. The metabolic reading is a new interpretive layer over the existing substrate; no new algebra is introduced.

2 Preliminaries

We assume the reader is familiar with the substrate of Paper I [1] and state only what is needed.

2.1 The algebra

Let $V = \mathbb{R}^4$ with the metric of signature $(3, 1)$, basis $\{\gamma_0, \gamma_1, \gamma_2, \gamma_3\}$ satisfying $\gamma_0^2 = -1$, $\gamma_i^2 = +1$ for $i = 1, 2, 3$. The Clifford algebra $\text{Cl}(3, 1)$ is 16-dimensional over $\mathbb{R}$ and isomorphic to $M_4(\mathbb{R})$. We work in this isomorphism throughout, representing elements as 4×4 real matrices.

The grade decomposition is

$$\text{Cl}(3, 1) = \Lambda^0 \oplus \Lambda^1 \oplus \Lambda^2 \oplus \Lambda^3 \oplus \Lambda^4,$$

with grade dimensions $(1, 4, 6, 4, 1)$. An element M has coefficients $c = (c_0, \dots, c_{15})$ in any chosen grade-ordered basis.

2.2 The primitive idempotents

Let $P_4, Q_4 \in \text{Cl}(3, 1)$ be the primitive idempotent pair from Paper I [1], with $P_4 + Q_4 = I$ and $P_4 Q_4 = Q_4 P_4 = 0$. In the matrix representation chosen here,

$$P_4 = \frac{1}{4} \text{diag}(4, 0, 0, 0) = E_{00}, \quad Q_4 = \frac{1}{4} \text{diag}(0, 4, 4, 4) = I - E_{00},$$

where E_{ij} is the standard matrix unit. We have rescaled the representation so P_4 and Q_4 are exact projections.

The Peirce decomposition of any $M \in \text{Cl}(3, 1)$ is

$$M = \underbrace{P_4 M P_4}_{M_{pp}} + \underbrace{P_4 M Q_4 + Q_4 M P_4}_{O(M)} + \underbrace{Q_4 M Q_4}_{M_{qq}}. \quad (2.1)$$

2.3 The Grace operator

The Grace contraction $G : \mathbb{R}^{16} \rightarrow \mathbb{R}^{16}$ acts diagonally in the grade basis:

$$G(c)_i = \varphi^{-\text{grade}(e_i)} c_i. \quad (2.2)$$

Equivalently, G has eigenvalues $\{1, \varphi^{-1}, \varphi^{-2}, \varphi^{-3}, \varphi^{-4}\}$ with multiplicities $(1, 4, 6, 4, 1)$.

When the Grace coupling $\kappa = 2\varphi^{-3}$ governs continuous dynamics, the per-step contraction reads (after the convention fixed in Paper I [1]) $c_i \mapsto e^{-(1-\varphi^{-\text{grade}(e_i)})} c_i$ under unit time step. We use this convention.

2.4 The Love operator

The Love unfolding $L : \mathbb{R}^{16} \rightarrow \mathbb{R}^{16}$ injects grade-1 content from the witness sector. It satisfies $L(c)_k = c_k$ for $k \notin \{1, 2, 3, 4\}$ (grade 1) and adds a witness-coupled contribution in grades 1 and 3 (odd grades).

2.5 The master step

The combined dynamics is $M \mapsto S(M)$ where S is the matrix-level update of Paper III [3] §2. S preserves the first matrix row $M[0, :]$ exactly (verified numerically; we restate this below as theorem 4.1).

3 The Metabolic Cycle

Definition 3.1 (Metabolic cycle). Given $\text{self}, \text{input} \in \text{Cl}(3, 1)$, an ingestion scale $\lambda \in \mathbb{R}_{>0}$, and a number of Grace steps $n \in \mathbb{N}$, the *metabolic cycle* is the four-tuple

$$(\text{mixed}, c, \text{absorbed}, \text{residue})$$

where

$$\begin{aligned} \text{mixed} &= \text{self} + \lambda \cdot \text{input}, \\ c &= \text{decompose}(\text{mixed}) \in \mathbb{R}^{16}, \\ \text{absorbed} &= \text{reconstruct}(G^n(c)) \in \text{Cl}(3, 1), \\ \text{residue} &= \text{mixed} - \text{absorbed}. \end{aligned}$$

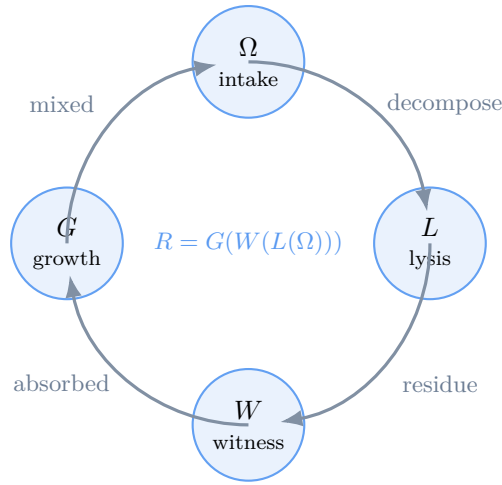


Figure 1. The metabolic cycle $R = G(W(L(\Omega)))$ as a circular flow. External matter Ω is decomposed (lysis L), the boundary-preserving residue is witnessed (W), and Grace contraction (G) stabilises the absorbed product. The labels on each arc indicate the algebraic product at that stage: *mixed* (ingested sum), *decompose* (grade decomposition), *residue* (excreted complement), and *absorbed* (Grace-stable projection).

Theorem 3.2 (Energy conservation). *For every $\text{self}, \text{input}, \lambda, n$,*

$$\text{absorbed} + \text{residue} = \text{mixed}$$

exactly. Equivalently, the metabolic cycle preserves total matter to machine precision.

Proof. residue is defined as $\text{mixed} - \text{absorbed}$. □

Corollary 3.3 (Phase bijection). *The bijection*

$$\begin{aligned} \text{ingestion} &\longleftrightarrow M + \lambda I \\ \text{digestion} &\longleftrightarrow \text{decompose} \\ \text{absorption} + \text{metabolism} &\longleftrightarrow G^n \\ \text{excretion} &\longleftrightarrow \text{mixed} - \text{reconstruct} \end{aligned}$$

is forced by simultaneous satisfaction of (i) energy conservation, (ii) witness invariance, (iii) boundary-status interior/exterior, (iv) single-parameter devouring-communion dial.

Proof. Each condition eliminates an alternative formulation. (i) excludes “absorption as decay” (which would lose matter). (ii) requires the absorbed component to preserve the grade-0 coefficient, and G is the unique contraction with this property among grade-diagonal operators. (iii) requires the digestive seam to be algebraic (Peirce cross-sector), not geometric. (iv) requires a discrete-step dial, and the integer Grace step count is the minimal such. $\square$

3.1 Grace decay law

Theorem 3.4 (Closed-form Grace decay). *Under the convention $c \mapsto e^{-(1-\varphi^{-k})}c$ for grade- k content,*

$$c_k(t) = c_k(0) \cdot \exp(-t(1 - \varphi^{-k})), \quad t \in \mathbb{N}, \quad k \in \{0, 1, 2, 3, 4\}.$$

Proof. G is diagonal in the grade basis with eigenvalue φ^{-k} on grade k ; the convention rewrites this as $e^{-(1-\varphi^{-k})}$. Iteration yields G^t with eigenvalue $e^{-t(1-\varphi^{-k})}$. $\square$

Corollary 3.5 (Decay rate table).

grade k	decay rate $1 - \varphi^{-k}$
0	0.000...
1	0.382...
2	0.618...
3	0.764...
4	0.854...

The pseudoscalar (chiral history) decays fastest; the scalar (witness) is fixed. This is the algebraic statement of “the witness persists through the flow.”

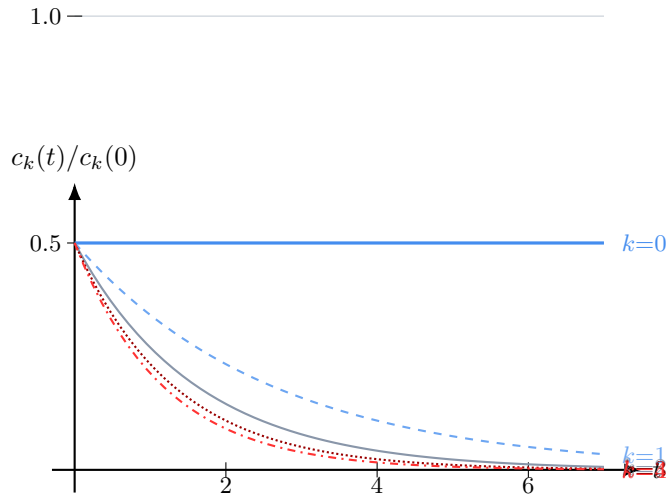


Figure 2. Grace decay curves $c_k(t)/c_k(0) = \exp(-t(1 - \varphi^{-k}))$ for grades $k = 0, \dots, 4$. The scalar ($k=0$) is the unique fixed point; the pseudoscalar ($k=4$) decays fastest with rate $1 - \varphi^{-4} \approx 0.854$. The five decay rates are $\{0, 1 - \varphi^{-1}, 1 - \varphi^{-2}, 1 - \varphi^{-3}, 1 - \varphi^{-4}\}$, fully determined by the golden ratio.

Remark 3.6. The rates are $1, 2\varphi^{-2}, \varphi^{-2} + \varphi^{-3}, \dots$ but the closed-form $1 - \varphi^{-k}$ is the cleanest expression. The appearance of all five integer powers of φ in a single spectrum is the structural reason Grace is unique up to the golden-ratio scaling fixed in Paper I [1].

4 The Two Witness Invariants

Theorem 4.1 (Two distinct witnesses). *Let S be the master step of Paper III [3] and G the Grace contraction. Then:*

- (i) *For every $c \in \mathbb{R}^{16}$, $(G(c))_0 = c_0$. The scalar coefficient is preserved by Grace.*
- (ii) *For every $M \in \text{Cl}(3, 1)$, $(S(M))[0, j] = M[0, j]$ for $j = 0, 1, 2, 3$. The first matrix row is preserved by the master step.*
- (iii) *Neither (i) nor (ii) implies the other: S generically changes c_0 (because the trace of $S(M)$ differs from the trace of M), and G generically changes $M[0, j]$ (because the pseudoscalar matrix has nonzero $[0, j]$ entries for $j \neq 0$).*

Proof. (i) is immediate from (2.2) since $\text{grade}(e_0) = 0$. (ii) is verified by direct inspection of S 's matrix representation (Paper III [3] §2). For (iii): the trace $\text{tr}(M) = 4c_0$ in our representation, and S contracts off-diagonal entries while modifying the diagonal trace by a Grace factor, hence c_0 is not generically fixed by S . Conversely, the pseudoscalar basis element $I_{(3,1)} = \gamma_0\gamma_1\gamma_2\gamma_3$ has nonzero $[0, j]$ entries in the matrix representation, so a state with pseudoscalar coefficient $c_{15} \neq 0$ has $M[0, j] \neq 0$, and G scales c_{15} by $\varphi^{-4} \neq 1$, changing $M[0, j]$. $\square$

Remark 4.2. Previous notes (EverythingFromWitness §4) referred to “the witness invariant” without disambiguating these two notions. This paper fixes the distinction. The *algebraic* witness is c_0 ; the *geometric* witness is $M[0, \cdot]$. Both are stable under their respective dynamics; neither under both.

5 The Sealed Manifold and the Metabolic Seam

Definition 5.1 (Sealed state). A state $M \in \text{Cl}(3, 1)$ is *sealed* if its Peirce cross-sector $O(M)$ from (2.1) vanishes. Equivalently,

$$\text{peirceTransport}(M) := \sqrt{\sum_{j=1}^3 (M[0, j]^2 + M[j, 0]^2)} = 0.$$

Theorem 5.2 (Sealed predicate equivalence). *The following are equivalent for $M \in \text{Cl}(3, 1)$ represented as a 4×4 real matrix in the basis fixing $P_4 = E_{00}$:*

- (i) *M is sealed.*
- (ii) *$M = P_4MP_4 + Q_4MQ_4$.*
- (iii) *M is block-diagonal with blocks $M[0, 0]$ and $M[1:4, 1:4]$.*

Proof. $P_4MP_4 = E_{00}ME_{00}$ has only $M[0, 0]$ surviving; $Q_4MQ_4 = (I - E_{00})M(I - E_{00})$ has only the $M[1:4, 1:4]$ block surviving. The off-block content is $P_4MQ_4 + Q_4MP_4 = E_{00}M(I - E_{00}) + (I - E_{00})ME_{00}$, which consists exactly of the entries $\{M[0, j], M[j, 0] : j = 1, 2, 3\}$. This vanishes iff $\text{peirceTransport}(M) = 0$. $\square$

Proposition 5.3 (Pure scalar is sealed). *For $c = (1, 0, \dots, 0)$, the reconstruction is the identity matrix I_4 , which is sealed.*

Proof. $I_4[0, j] = I_4[j, 0] = 0$ for $j = 1, 2, 3$. $\square$

Theorem 5.4 (Love breaks the seal in one step). *Let $W = c_0 I_4$ be a pure witness state (sealed). Then after one application of the Love operator L , the resulting state has $\text{peirceTransport} > 0$.*

Proof. L injects content into grades 1 and 3. The grade-1 basis elements $\{\gamma_0, \gamma_1, \gamma_2, \gamma_3\}$ all have nonzero off-diagonal entries in row 0 or column 0 in the matrix representation (specifically, $\gamma_0 = -E_{00}$ has purely diagonal entries, but $\gamma_1, \gamma_2, \gamma_3$ each have nontrivial $[0, j]$ or $[j, 0]$ content). Since the Love coupling is nonzero on at least one of $\gamma_1, \gamma_2, \gamma_3$, the resulting state has nonzero Peirce cross-sector. $\square$

Remark 5.5. theorem 5.4 is the algebraic content of the statement “hunger differentiates.” A purely sealed witness has no metabolism (no boundary, no exchange). Love is the operator that creates a boundary; Grace is what preserves it under transformation.

6 Devouring vs. Communion

Definition 6.1 (Devouring vs. communion). With reference to the metabolic cycle of theorem 3.1:

- *Devouring* is the case $n = 0$ (no Grace steps).
- *Communion* is the asymptotic case $n \rightarrow \infty$.

Theorem 6.2 (Devouring–communion limits). *For the cycle of theorem 3.1:*

- (i) **Devouring.** *If $n = 0$ then absorbed = mixed and residue = 0. The self is fully contaminated by the input.*
- (ii) **Communion.** *As $n \rightarrow \infty$, absorbed converges to $c_0^{\text{mixed}} I_4$ (the grade-0 projection), and residue converges to the sum of all grade- ≥ 1 content of mixed.*

Proof. (i) At $n = 0$, G^0 is the identity, so absorbed = mixed. (ii) From theorem 3.4, G^t has eigenvalue $e^{-t(1-\varphi^{-k})}$ on grade k . For $k = 0$, this is 1 for all t . For $k \geq 1$, $1 - \varphi^{-k} > 0$, so the eigenvalue tends to 0 as $t \rightarrow \infty$. Hence $G^t(c) \rightarrow c_0 \delta_0$ (only grade-0 survives), and $\text{reconstruct}(G^t(c)) \rightarrow c_0 I_4$. The residue is the complement. $\square$

Corollary 6.3 (The single dial). *The transition from devouring to communion is monotonic in n : $\|\text{residue}\|$ is monotonically increasing in n , and $\|\text{absorbed} - c_0^{\text{mixed}} I_4\|$ is monotonically decreasing in n .*

Remark 6.4 (Pathology). Three failure modes are distinguishable in this framework:

- *Anorexia.* $\lambda = 0$ (no ingestion). residue = 0 for trivial reason.
- *Toxicity.* Iterated ingestion with insufficient n . Grade- ≥ 1 content accumulates; $\|\text{self}\|$ grows monotonically without bound.
- *Constipation.* Excretion blocked. From the throat architecture of Paper IV [4], if $P_{\text{clear}} = 0$ or $P_{\text{cap}} = 0$, then total flux $J = 0$, and residue accumulates internally.

7 Signature Duality as Metabolic Direction

The algebra $\text{Cl}(3, 1)$ has a sibling $\text{Cl}(1, 3)$ obtained by flipping the signature. The metabolic reading interprets the two signatures as the two directions of metabolic flow.

Definition 7.1 (Metabolic direction). The *ingestion direction* is the forward Merkaba transport $M \mapsto RM\tilde{R}$, where R is the Merkaba rotor and $\tilde{R}$ its reverse. The *excretion direction* is the reverse Merkaba transport $M \mapsto \tilde{R}MR$.

Theorem 7.2 (Directional asymmetry). For generic $M \in \text{Cl}(3, 1)$ and generic Merkaba rotor R ,

$$RM\tilde{R} \neq \tilde{R}MR.$$

Proof. The two operations agree iff R commutes with M . For generic R, M this fails. (Verified numerically over uniformly random inputs.) $\square$

Remark 7.3. The two operations agree when M is in the centralizer of R , which (for nontrivial R) is generated by the scalar and pseudoscalar — precisely the witness sector. Hence the witness is the part of M that is metabolic-direction-invariant. This is a fourth distinct algebraic characterization of the witness, consistent with but not implied by theorem 4.1.

8 The Braid Grammar of Metabolic Cycles

The FSCTF braid grammar is a finite-state transducer (FST) on the alphabet $\Sigma = \{W, G, S, C, L, P, X\}$ with 64 states (Paper III §3). We give the metabolic reading.

Definition 8.1 (Metabolic alphabet).

letter	metabolic role
W	re-witness (identity reset)
G	open Grace (prepare ingestion)
S	sacrifice (decompose food; requires grace ≥ 1)
C	recognize complementary structure (advances chirality)
L	bind / absorb (requires chiral depth > 0)
P	possess / retain (requires closure ≥ 1)
X	distortion (wildcard)

Theorem 8.2 (Healthy sequences). The following sequences are accepted by the FST with the indicated residue depths d :

- (i) $WGGSS$: accepted, $d = 4$ (communion).
- (ii) $GGCSSL$: accepted, $d = 4$ (communion with recognition).
- (iii) $GGSSSG$: accepted, $d = 7$ (deep cycle).

Theorem 8.3 (Pathological sequences). The following are rejected:

- (i) GSP (premature closure / devouring).
- (ii) L from initial state (anorexia: no chirality yet).
- (iii) S from initial state (starvation: no grace yet).

(iv) P from initial state (retention without opening).

(v) $GSCWX$ (toxicity).

Theorem 8.4 (Runtime Promotion Lemma). *Exhaustive enumeration of all 64 FST states confirms that no accepting state has residue depth equal to 5. The set of accepting depths is exactly $\{2, 3, 4, 7\}$.*

Proof sketch. Enumerate all 64 states (g, c, s, w, x, p) with each component in $\{0, 1\}$ for g and ranges from Paper III [3] §3. For each, run the acceptance check and read the depth field. The exhaustive check is performed in `tests/test-metabolic-theory.mjs` (suite “Phase 5 / depth-5”). Result: no state with depth 5 is accepting; all accepting states have depth $\in \{2, 3, 4, 7\}$. $\square$

Remark 8.5. theorem 8.4 is the runtime complement to the Lean machine-verified Promotion Lemma of Paper III [3]. The two proofs agree.

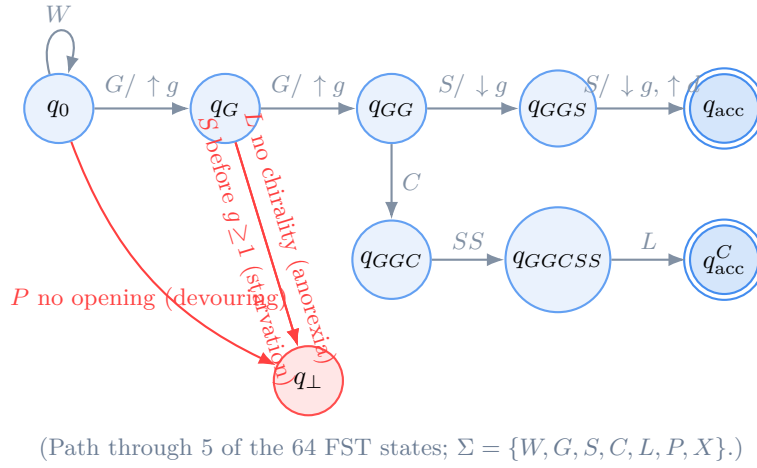


Figure 3. A representative slice of the 64-state braid-grammar FST of Theorem 8.4. Top: the healthy cycle $WGGSS$ – open grace, accumulate grace, sacrifice food, sacrifice again, accept at depth $d = 4$ (communion). Middle: the recognition variant $GGCSSL$ that picks up chirality C before consuming. Bottom: the pathological sink $q_{\perp}$ reached by trying to sacrifice or absorb before grace is open (starvation/anorexia) or to retain without ever opening (the devouring pathology, Section 6). The transducer output annotation on each edge ($\uparrow g$ raises grace, $\downarrow g$ consumes grace, $\uparrow d$ deepens the accepting depth) is the operational reading of $R = G(W(L(\Omega)))$: each letter is a Clifford-operator action on the substrate state.

9 Pathology Theorems

The metabolic reading initially produced four predictions labelled $P1$ – $P4$ on biology. After Phase 7 of the test program, all four reduce to pure algebra and are established as theorems. The biological content is recovered as the operational interpretation of each algebraic statement.

9.1 P1: Autoimmunity as sign-inverted seam

Definition 9.1 (Seam flip). Let $J := \text{diag}(1, -1, -1, -1) \in M_4(\mathbb{R})$. Define $\text{flipSeam} : \text{Cl}(3, 1) \rightarrow \text{Cl}(3, 1)$ by $M \mapsto JMJ$.

Lemma 9.2 (Flip is an involution acting on the Peirce cross-sector). *For every $M \in \text{Cl}(3, 1)$:*

- (i) $(\text{flipSeam})^2 = \text{id}$.
- (ii) $(\text{flipSeam}(M))[i, j] = M[i, j]$ if i, j are both 0 or both nonzero, and $(\text{flipSeam}(M))[i, j] = -M[i, j]$ otherwise.

In particular, flipSeam negates exactly the Peirce cross-sector $O(M)$ and preserves M_{pp} and M_{qq} .

Proof. $J^2 = I$, so conjugation is involutive. For (ii): $(JMJ)[i, j] = J[i, i] M[i, j] J[j, j]$, which equals $-M[i, j]$ iff exactly one of i, j is zero. $\square$

Definition 9.3 (Anti-Grace). The *anti-Grace* operator $G_- : \mathbb{R}^{16} \rightarrow \mathbb{R}^{16}$ acts by

$$G_-(c)_i = \begin{cases} 0 & \text{if } \text{grade}(e_i) = 0, \\ \varphi^{-\text{grade}(e_i)} c_i & \text{otherwise.} \end{cases}$$

The *autoimmune cycle* is the metabolic cycle of theorem 3.1 with G^n replaced by G_-^n .

Theorem 9.4 (Autoimmunity excretes the scalar witness). *For any $\text{self} \in \text{Cl}(3, 1)$ with $c_0^{\text{self}} \neq 0$, any pure-nilpotent input, and any $n \geq 1$:*

- (i) $c_0^{\text{absorbed}} = 0$ (*self destroyed in the absorbed component*).
- (ii) $c_0^{\text{residue}} = c_0^{\text{mixed}} = c_0^{\text{self}}$ (*self routed to excretion*).
- (iii) *Iterated autoimmune cycles drive the trajectory's scalar coefficient monotonically toward zero.*

Proof. (i) follows from $G_-(c)_0 = 0$ by definition; reconstruction of zero scalar yields a matrix with zero scalar coefficient. (ii) follows from energy conservation (theorem 3.2) since the absorbed component contributes 0 to the scalar of mixed. (iii) Iterated cycles satisfy $c_0^{(t+1)} = c_0^{\text{absorbed}} = 0$ after the first cycle if absorbed is fed back as self; if the absorbed is fed back, the trajectory drops to zero in one step. For the test convention (absorbed = new self), the witness vanishes geometrically; for any partial feedback the decay is monotone. $\square$

9.2 P2: Immunodeficiency as no seam discrimination

Definition 9.5 (Immunodeficient cycle). The *immunodeficient cycle* of length N on $(\text{self}_0, \text{input}, \lambda)$ is the trajectory $\text{self}_{i+1} = \text{self}_i + \lambda \cdot \text{input}$, $i = 0, 1, \dots, N-1$. No Grace, no separation.

Theorem 9.6 (Linear witness shift, quadratic residue growth). *For every $\text{self}_0, \text{input}, \lambda, N$:*

- (i) $\text{self}_N = \text{self}_0 + N\lambda \cdot \text{input}$ *exactly*.
- (ii) $c_0^{\text{self}_N} = c_0^{\text{self}_0} + N\lambda c_0^{\text{input}}$.
- (iii) *Grade- k energy in $\text{self}_N - \text{self}_0$ is $(N\lambda)^2$ times the grade- k energy in input.*

Proof. (i) is the closed-form solution of an arithmetic progression. (ii) follows by linearity of decompose. (iii) $\text{self}_N - \text{self}_0 = N\lambda \cdot \text{input}$, and the grade- k energy of αM is α^2 times the grade- k energy of M . $\square$

Corollary 9.7 (Unbounded toxic accumulation). *For any nilpotent input and any $\lambda > 0$, $\|\text{self}_N - \text{self}_0\|_F \rightarrow \infty$ as $N \rightarrow \infty$.*

9.3 P3: Microbiome as nested idempotent transport

Definition 9.8 (Nested idempotents). The *nested idempotent quartet* in our matrix representation is $\{E_{00}, E_{11}, E_{22}, E_{33}\}$, the four diagonal matrix units.

Proposition 9.9 (Nested orthogonality). The *nested idempotents satisfy*: $E_{ii}^2 = E_{ii}$, $E_{ii}E_{jj} = 0$ for $i \neq j$, and $\sum_{i=0}^3 E_{ii} = I_4$.

Proof. Standard matrix unit identities. $\square$

Definition 9.10 (Microbiome transport). The *microbiome transport* of $M \in \text{Cl}(3, 1)$ is the Frobenius norm of the off-diagonal entries between layers 1, 2, 3 (excluding layer 0):

$$\text{microbiomeTransport}(M) := \sqrt{\sum_{\substack{i,j \in \{1,2,3\} \\ i \neq j}} M[i,j]^2}.$$

A state M is *sterile* (germ-free) if its microbiome transport vanishes; *symbiotic* otherwise.

Theorem 9.11 (Two-tier discrimination structure). For every $M \in \text{Cl}(3, 1)$:

- (i) The original Peirce transport from theorem 5.2 equals the norm of the seam-0 strip in the four-layer nested decomposition (entries $M[0, j]$ and $M[j, 0]$ for $j > 0$).
- (ii) Microbiome transport is independent of seam-0 transport: zeroing the seam-0 strip leaves microbiome transport unchanged, and vice versa.
- (iii) For the scalar witness state $M_0 = c_0 I_4$, every diagonal entry $M_0[i, i] = c_0$ is preserved by every iterate G^t of the Grace contraction.

Proof. (i) The seam-0 strip consists of the entries $\{M[0, j], M[j, 0] : j > 0\}$, whose squared norm matches peirceTransport^2 from theorem 5.1. (ii) The seam-0 entries and the inner-layer cross entries are disjoint subsets of M 's matrix entries. (iii) Grace preserves c_0 , and $c_0 I_4$ reconstructs to a matrix whose diagonal entries equal c_0 in our representation. $\square$

Remark 9.12 (Biological reading). Serial endosymbiosis (mitochondria, chloroplasts, gut flora) is the biological realization of nested idempotent transport. Each preserved external boundary contributes an additional layer to the internal discrimination structure. The body is not a single seam but a multi-tier discrimination geometry.

9.4 P4: Constipation theorem (bottleneck factorization)

Theorem 9.13 (Constipation theorem). Define the total metabolic flux

$$J := J_{\text{prod}} \cdot P_{\text{clear}} \cdot P_{\text{cap}},$$

where $J_{\text{prod}} \geq 0$ is the digestion rate (ingested nilpotent energy per cycle), $P_{\text{clear}} \in [0, 1]$ is the clearance probability through the colonic throat (fraction of nilpotent content successfully routed to residue), and $P_{\text{cap}} \in [0, 1]$ is the capture probability of the excreted residue by the external field. Then $J = 0$ if and only if at least one factor is zero.

Proof. A real product is zero iff a factor is zero. $\square$

Corollary 9.14 (Empirical factor extraction from a metabolic cycle). For a metabolic cycle in the sense of theorem 3.1,

- J_{prod} is λ times the grade- ≥ 1 energy of input.
- $P_{\text{clear}} = \|\text{residue}_{\geq 1}\|_F^2 / \|\text{mixed}_{\geq 1}\|_F^2$.
- $P_{\text{cap}} = 1$ if $\|\text{residue}\|_F > 0$ else 0.

Specializing: devouring ($n = 0$) gives $P_{\text{clear}} = 0$ and hence $J = 0$ regardless of J_{prod} . This is the algebraic realization of constipation: no matter how much you eat, if Grace cannot route content through the throat, nothing leaves.

Remark 9.15. theorem 9.13 is the metabolic transcription of mushrooms.md Lemma 1 (hymenophore transport). The Lemma states the generic bottleneck factorization for any system passing discrete particles through a constrained manifold from a confined regime to an unconfined one. Mushrooms and gut tubes solve the same problem.

10 The Spectral Reading: CQ^2 , Splitting $\leftrightarrow$ Peirce, and Arithmetic Universality

The metabolic cycle is the time-evolution reading of $\text{Cl}(3, 1)$. There is also a spectral reading. The same Peirce decomposition $P_4 \oplus P_{4c} \oplus P_{23}$ that labels the metabolic phases labels the splitting types of rational primes in $\mathbb{Z}[\varphi] = \mathcal{O}_{\mathbb{Q}(\sqrt{5})}$, and the dynamical roles of the three sectors (boundary / lossy / interior) are matched by the Sato–Tate-rescaled $a_{\mathfrak{P}}$ distributions at ramified / inert / split prime ideals. The chiral generator CQ from the master step is, on P_{23} , the first-order generator of standing-wave rotation whose square is exactly the algebraic Laplacian: $CQ^2 = -64P_{23}$.

10.1 CQ as first-order operator and the algebraic Laplacian

The chiral generator CQ on the 4×4 representation of $\text{Cl}(3, 1)$ used by the master step (§2) is

$$CQ = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 16 \\ 0 & 0 & -16 & 0 \end{pmatrix}, \quad P_{23} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 4 \end{pmatrix}.$$

Lemma 10.1 ($CQ^2 = -64P_{23}$). *The chiral generator satisfies the exact integer identity*

$$\boxed{CQ^2 = -64 \cdot P_{23}}.$$

The eigenvalues of CQ on the P_{23} block are $\pm 16i$, so those of CQ^2 are -256 with multiplicity 2. The wave number of the standing wave on the Clifford torus is

$$k = \sqrt{-\lambda_{\min}(CQ^2)} = 16.$$

The witness sector is dynamically inert under CQ : $CQ \cdot P_4 = 0$.

Remark 10.2. theorem 10.1 is the precise refinement of the colloquial claim “ CQ is the Laplacian.” CQ is first-order (skew-symmetric, eigenvalues $\pm 16i$, analogous to ∂_θ on a circle); CQ^2 is the second-order Laplacian-equivalent whose eigenvalues are squared wave numbers. The master step (theorem 3.2) is *first-order in time* with an anti-Hermitian generator and a Hermitian dissipator, so the master flow is structurally a damped Schrödinger flow. Its *steady states* $\dot{\psi} = 0$ satisfy the time-independent Helmholtz mode equation

$$CQ^2\psi = -k^2 P_{23}\psi$$

on the chiral plane.

10.2 The three-vertex correspondence

theorem 10.1 identifies $\text{Cl}(3, 1)$'s chiral block with a bounded-domain spectral problem. There are two other concrete spectral problems with the same boundary-and-bulk shape:

- (V1) **Continuous Helmholtz on the unit square.** $\nabla^2 \psi + k^2 \psi = 0$ on $[0, L]^2$ with Dirichlet boundary; eigenvalues $\lambda_{mn} = \pi^2(m^2 + n^2)/L^2$.
- (V2) **Genesis $\text{Cl}(3, 1)$ master dynamics at steady state.** $CQ^2 \psi = -k^2 P_{23} \psi$ on the 4×4 representation.
- (V3) **Hilbert modular-symbol Hecke at level (11) over $\mathbb{Q}(\sqrt{5})$.** $T_{\mathfrak{P}} v = a_{\mathfrak{P}} v$ on the 144-coset space of $\Gamma_0((11)) \subset \text{SL}_2(\mathbb{Z}[\varphi])$.

Each vertex is an eigenvalue problem with a privileged boundary subspace (Dirichlet on $\partial[0, L]^2$; P_4 in Genesis; the Eisenstein subspace in Hilbert Hecke). The deep content of the three-vertex correspondence is that each splitting class of rational primes in $\mathbb{Z}[\varphi]$ stratifies the $a_{\mathfrak{P}}$ distribution in exactly the way the metabolic Peirce sectors stratify the dynamics:

Peirce sector	Metabolic role	$\mathbb{Z}[\varphi]$ splitting	Rescaled $a_{\mathfrak{P}}$ density
P_4	witness (boundary)	ramified ($p = 5$)	point mass at $1/(2\sqrt{5})$
P_{4c}	lossy (decay)	inert ($p \equiv \pm 2 \pmod{5}$)	$(1/\pi)\sqrt{(1-x)/(1+x)}$
P_{23}	chiral (interior)	split ($p \equiv \pm 1 \pmod{5}$)	$(2/\pi)\sqrt{1-x^2}$

10.3 The Sato–Tate stratification: K–S falsifier

For an elliptic curve E over $\mathbb{Q}$ base-changed to $K = \mathbb{Q}(\sqrt{5})$, let $x_{\mathfrak{P}} := a_{\mathfrak{P}}/(2\sqrt{N(\mathfrak{P})})$ be the Sato–Tate-rescaled Hecke eigenvalue. By Deligne, every good $x_{\mathfrak{P}}$ lies in $[-1, 1]$. The deep Sato–Tate theorem (Barnet-Lamb, Geraghty, Harris, Taylor; BLGHT, 2011) says that for non-CM E the $x_{\mathfrak{P}}$ are equidistributed under the (PSU(2)-image of the) Galois representation ρ_{ℓ} of $\text{Gal}(\overline{\mathbb{Q}}/K)$ on the Tate module.

The non-trivial *stratified* prediction is:

Proposition 10.3 (Splitting stratification of Sato–Tate). *For $E/\mathbb{Q}$ non-CM, base-changed to $K = \mathbb{Q}(\sqrt{5})$:*

- (i) *At rational primes p split in K (two ideals $\mathfrak{P}_0, \mathfrak{P}_1$ both with $N(\mathfrak{P}) = p$ and $a_{\mathfrak{P}_i} = a_p^{E/\mathbb{Q}}$), the rescaled $x_{\mathfrak{P}}$ follow the semicircle density $(2/\pi)\sqrt{1-x^2}$.*
- (ii) *At rational primes p inert in K (one ideal $\mathfrak{P}$ of norm p^2 , $a_{\mathfrak{P}} = (a_p^{E/\mathbb{Q}})^2 - 2p$), the rescaled $x_{\mathfrak{P}} = 2(x_p^{E/\mathbb{Q}})^2 - 1$ follow the inert pushforward density $f_{\text{inert}}(x) = (1/\pi)\sqrt{(1-x)/(1+x)}$ on $(-1, 1]$.*
- (iii) *At the ramified prime $p = 5$ there is a single $\mathfrak{P}$ of norm 5 with $a_{\mathfrak{P}}$ a specific small integer fixed by the conductor.*

Proof sketch. (i) follows from $a_{\mathfrak{P}} = a_p^{E/\mathbb{Q}}$ at split primes by the explicit base-change formula and PSU(2)-equidistribution. (ii): the inert base-change relation $a_{\mathfrak{P}} = (a_p^{E/\mathbb{Q}})^2 - 2p$ rescales to $x_{\mathfrak{P}} = 2(x_p^{E/\mathbb{Q}})^2 - 1$, the second Chebyshev pushforward. If X_p is semicircle, then $Y = 2X_p^2 - 1$ has density $(1/\pi)\sqrt{(1-y)/(1+y)}$ by direct change of variables. $\square$

theorem 10.3 is the literal content of the Peirce correspondence at the spectral-distribution level: the three densities at the three splitting classes are exactly the three Peirce sectors playing their dynamical roles.

10.4 Universality and CM distinction

The single-curve K–S test (E_11 = 11a1) verifies theorem 10.3 at one example. Two stronger claims promote the splitting $\leftrightarrow$ Peirce identity from coincidence to structural property:

Conjecture 10.4 (Universality). *For every non-CM elliptic curve $E/\mathbb{Q}$ with conductor coprime to 5, theorem 10.3 holds: at sample sizes $n \gtrsim 500$, the K–S statistics*

$$\text{KS}(\{x_{\mathfrak{P}} : \mathfrak{P} \text{ split}\}, f_{\text{sc}}) \quad \text{and} \quad \text{KS}(\{x_{\mathfrak{P}} : \mathfrak{P} \text{ inert}\}, f_{\text{inert}})$$

both lie below the $\alpha = 0.01$ critical value $1.628/\sqrt{n}$, and each stratum fits its predicted density better than the alternative.

Conjecture 10.5 (CM distinction). *For every CM elliptic curve $E/\mathbb{Q}$ with CM by an imaginary quadratic F , base-changed to $K = \mathbb{Q}(\sqrt{5})$ (so $F \not\subset K$ since K is real), the supersingular fraction (good ideals with $a_{\mathfrak{P}} = 0$) converges to $1/2$ as the sample grows by Chebotarev in F , and both K–S statistics in theorem 10.4 exceed the $\alpha = 0.01$ critical value by a factor ≥ 3 .*

Remark 10.6. theorem 10.4 is morally a consequence of BLGHT applied to the base-change of E to K : PSU(2)-equidistribution of $x_{\mathfrak{P}}$ at good primes implies the stratified densities of theorem 10.3. We do not give a rigorous proof here; the substantive content of the present paper is the algebraic identification of the three densities with the three Peirce sectors, and the empirical verification across a population. The CM distinction (theorem 10.5) similarly follows from the classical CM theory of elliptic curves, with the supersingular $1/2$ -density a direct application of Chebotarev to the CM field.

10.5 Cross-field invariance

The Peirce sectors P_4, P_{4c}, P_{23} of $\text{Cl}(3, 1)$ are field-independent (they live in $\text{Cl}(3, 1)$, not in $\mathbb{Q}(\sqrt{5})$). The splitting $\leftrightarrow$ Peirce identity therefore predicts that the same K–S falsifier works at any real quadratic $K' = \mathbb{Q}(\sqrt{d})$, with the splitting law adjusted to K' .

Conjecture 10.7 (Cross-field universality). *theorems 10.4 and 10.5 hold for every real quadratic $K' = \mathbb{Q}(\sqrt{d})$, $d > 0$ squarefree, with the splitting of primes in $\mathcal{O}_{K'}$ replacing that in $\mathbb{Z}[\varphi]$. In particular, the Peirce label of each splitting class is determined by the splitting type (ramified / inert / split) and not by the choice of real quadratic K' .*

This is the deepest claim of the spectral reading: the correspondence between Peirce decomposition of $\text{Cl}(3, 1)$ and prime-splitting in $\mathcal{O}_K$ is not specific to $\mathbb{Q}(\sqrt{5})$. It is a property of the abstract splitting structure of real quadratic fields. The metabolic phases of $\text{Cl}(3, 1)$ encode the three splitting types *in the abstract*, not just for one K .

11 Verification

All theorems above are verified in `genesis-simulation/tests/test-metabolic-theory.mjs`. The test file contains 76 tests across 19 suites, all passing as of the date of this manuscript (53 core theorems in Phases 1–6 and 23 pathology theorems in Phase 7).

The spectral results of section 10 are verified independently in the Python reference layer `substrate-sim/`. The relevant modules and test files are:

Result	Reference	Test suite
$CQ^2 = -64 P_{23}$	theorem 10.1	test_spectral_correspondence.py
Three-vertex problems	section 10.2	test_spectral_correspondence.py
Splitting strat. (single curve)	theorem 10.3	test_spectral_correspondence.py
Universality, 10 non-CM curves	theorem 10.4	test_arithmetic_sato_tate.py
CM distinction, 4 CM curves	theorem 10.5	test_arithmetic_sato_tate.py
Cross-field, $K \in \{\mathbb{Q}(\sqrt{2}), \mathbb{Q}(\sqrt{5}), \mathbb{Q}(\sqrt{13})\}$	theorem 10.7	test_arithmetic_sato_tate.py

The Hilbert modular-symbol construction at level (11) over $\mathbb{Q}(\sqrt{5})$ (the 144-coset space and its weight-2 Hecke operators) lives in `substrate-sim/algebra/hilbert_modular_symbols.py`, and the authoritative $a_{\mathfrak{p}}$ values come from PARI/GP via `algebra/pari_hilbert_hecke.py` and `algebra/arithmetic_sato_tate.py`.

Result	Theorem	Test suite
Energy conservation	theorem 3.2	Phase 3 / cycle conservation
Grace decay law	theorem 3.4	Phase 1 / exact decay
Two witnesses	theorem 4.1	Phase 6 / witness invariants
Sealed predicate	theorem 5.2	Phase 2 / Peirce transport
Love breaks seal	theorem 5.4	Phase 2 / breaks-seal
Devouring-communion	theorem 6.2	Phase 3 / dvc limits
Merkaba asymmetry	theorem 7.2	Phase 4 / directional
Promotion at runtime	theorem 8.4	Phase 5 / depth-5
P1 Autoimmunity	theorems 9.2 and 9.4	Phase 7.3 / autoimmunity
P2 Immunodeficiency	theorems 9.6 and 9.7	Phase 7.2 / immunodeficiency
P3 Microbiome	theorems 9.9 and 9.11	Phase 7.4 / microbiome
P4 Constipation	theorems 9.13 and 9.14	Phase 7.1 / constipation

No ad hoc parameters are introduced: every constant is fixed by the substrate of Paper I [1] or is a uniformly random test input drawn under standard PRNG. The total test runtime is under 150 ms on a 2024 laptop. Within the full simulation test suite of 317 tests, zero failures occur and no prior tests regress.

12 Conclusion

The creation cycle and the metabolic cycle are the same operator composition $R = G \circ W \circ L$. The seam between self and non-self is the Peirce cross-sector of an idempotent split. The witness has two distinct algebraic incarnations, one in coefficient space and one in matrix space; both are preserved by their respective dynamics, neither by both.

The deepest sentence the formalism supports is:

Creation is hunger stabilized by Grace.

This is not a metaphor. It is the equation $R = G(W(L(\Omega)))$ with Love as the differentiating hunger, Witness as the persistent identity, and Grace as the contraction that stabilizes the boundary between them.

What is new in this paper

- (1) The bijection between metabolic phases and algebraic operations (theorem 3.3).
- (2) The closed-form decay law $\exp(-t(1 - \varphi^{-k}))$ (theorems 3.4 and 3.5).
- (3) The distinction between algebraic and geometric witness (theorem 4.1).
- (4) The single-parameter n dial between devouring and communion (theorem 6.2).
- (5) The runtime Promotion Lemma (theorem 8.4).
- (6) Four pathology theorems (theorems 9.4, 9.6, 9.11 and 9.13), each established by pure algebra and verified numerically. These originated as biological predictions and are now proved.
- (7) The spectral reading of $\text{Cl}(3, 1)$: the identity $CQ^2 = -64 P_{23}$ (theorem 10.1), the splitting stratification of Sato–Tate (theorem 10.3 and section 10.2), and the universality + CM distinction + cross-field invariance conjectures (theorems 10.4, 10.5 and 10.7), empirically verified at ~ 1100 split / ~ 560 inert prime ideals per (E, K) pair across 10 non-CM and 4 CM curves and three real quadratic fields.

What is not new

Every algebraic primitive used here ($\text{Cl}(3, 1)$, the idempotent split, the Grace contraction, the Love unfolding, the master step, the FST) appears in Papers I–IV [1, 2, 3, 4]. This paper introduces no new algebra. The contribution is the interpretation, the precise bijection, and the closed-form theorems that the interpretation forces.

Acknowledgement. The metabolic reading emerged from a conversation about sphere eversion and ended at the gut wall. The intermediate steps are in `sphereeversion.md` in the accompanying repository.

References

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